197. Rising Temperature

Easy | Topics | Companies

SQL Schema > Pandas Schema >

Table: `Weather`

```
+---------------+---------+
| Column Name   | Type    |
+---------------+---------+
| id            | int     |
| recordDate    | date    |
| temperature   | int     |
+---------------+---------+
id is the column with unique values for this table.
There are no different rows with the same recordDate.
This table contains information about the temperature on a certain day.
```

Write a solution to find all dates' `id` with higher temperatures compared to its previous calendar dates (yesterday).

Return the result table in **any order**.

The result format is in the following example.

Example 1:

Input:

Weather table:

id	recordDate	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

Output:

id
2
4

Explanation:

In 2015-01-02, the temperature was higher than the previous day (10 -> 25).
In 2015-01-04, the temperature was higher than the previous day (20 -> 30).

RS | Day-6 | 3/11/26

Leetcode-197: Rising temperature

We are given a table called Weather.
It contains 3 columns:

Column	Meaning
id	unique ID for each record
record_date	date when temp was recorded
temperature	temp. on that day.

Our task is:-
Find the IDs of all days when the temp was higher than the temp on the previous calender day (yesterday).

In simple:
- compare today's temp with yesterday's temperature
- if today's temp is higher, return today's ID.
- otherwise ignore it.

A We need to compare consecutive calender days, not simply consecutive rows.

Example: Input:

id	record_date	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

let's compare the temperatures:-

today	today's temp	yesterday's temp	higher
Jan2	25	10	y
Jan3	20	25	n
Jan4	30	20	y.

therefore,

output:- id:
2
4

sql concepts → self join

→ DATEDIFF() → it calculates the diff. between two dates in days.

syntax:- DATEDIFF(date1, date2)

ex DATEDIFF('2015-01-02', '2015-01-01')

o/p 1

SELECT w1.id → we retrieve the ID from which we represents today.

FROM Weather w1 → select today's & yesterday's records.

JOIN Weather w2 → we are joining the weather table with itself.

ON DATEDIFF(w1.RecordDate, w2.RecordDate)=1.

↳ today's date minus yesterday's date must equal 1 day.

WHERE w1.temperature > w2.temperature

↳ today's temperature must be greater than yesterday's temperature.

after self join & date matching:-

w.id	Today's date	today's temp	yesterday's temp
2	Jan2	25	10
3	Jan3	20	25
4	Jan4	30	20

required id: 2, 4

ID2: 25 > 10 → ✓

ID3: 20 > 25 → ✗

ID4: 30 > 20 → ✓.

An alternative approach

```
SELECT w1.id
FROM Weather w1
JOIN Weather w2
ON w2.RecordDate = DATE_SUB(w1.RecordDate,
                            INTERVAL 1 DAY)
WHERE w1.temperature > w2.temperature.
```

→ means subtract one day from today's date to get yesterday's date.

Accepted 15 / 15 testcases passed

Akankshyaa submitted at Nov 03, 2026 09:13

Solution

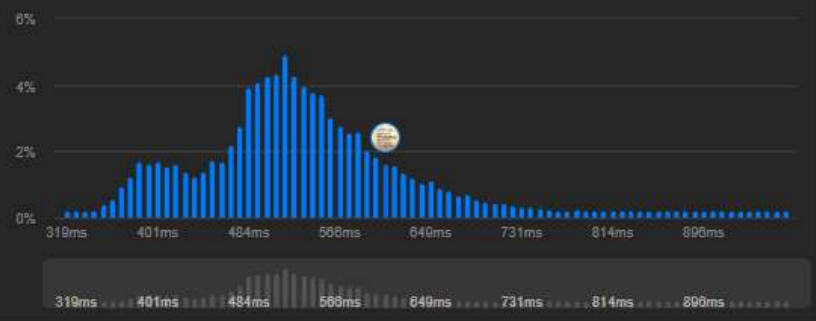
Save 41%

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Runtime

609 ms | Beats 22.42%



Code | MySQL

```mysql
# Write your MySQL query statement below
SELECT w1.id
FROM Weather w1
JOIN Weather w2
ON DATEDIFF(w1.recordDate,w2.recordDate)=1
WHERE w1.temperature>w2.temperature;
```

More challenges

- 180. Consecutive Numbers
- 2978. Symmetric Coordinates
- 3374. First Letter Capitalization II

```mysql
# Write your MySQL query statement below
SELECT w1.id
FROM Weather w1
JOIN Weather w2
ON DATEDIFF(w1.recordDate,w2.recordDate)=
WHERE w1.temperature>w2.temperature;
```

Saved

Testcase | Test Result

Accepted Runtime: 104 ms

Case 1

Input

Weather =

id	recordDate	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

Output

id
2
4

Expected

Id
2
4

Accepted 15 / 15 testcases passed

Akankshyaa submitted at Nov 03, 2026 09:15

Solution

Save 41%

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Runtime

536 ms | Beats 47.15%



Code | MySQL

```mysql
# Write your MySQL query statement below
SELECT w1.id
FROM Weather w1
JOIN Weather w2
ON w2.recordDate= DATE_SUB(w1.recordDate, INTERVAL 1 DAY)
WHERE w1.temperature>w2.temperature;
```

More challenges

- 180. Consecutive Numbers
- 2978. Symmetric Coordinates
- 3374. First Letter Capitalization II

```mysql
# Write your MySQL query statement below
SELECT w1.id
FROM Weather w1
JOIN Weather w2
ON w2.recordDate= DATE_SUB(w1.recordDate, INTERVAL 1 DAY)
WHERE w1.temperature>w2.temperature;
```

Saved

Testcase | Test Result

Accepted Runtime: 84 ms

Case 1

Input

Weather =

id	recordDate	temperature
1	2015-01-01	10
2	2015-01-02	25
3	2015-01-03	20
4	2015-01-04	30

Output

id
2
4

Expected

Id
2
4